



## SEMESTER END EXAMINATIONS - APRIL 2023

**Program : B.E. : Common to all Programs**

**Semester : III**

**Course Name : Unix Shell Programming**

**Max. Marks : 50**

**Course Code : CSAEC310**

**Duration : 2 Hrs**

### Instructions to the Candidates:

- PART-A** Answer all the questions.
- PART-B** Answer one full question from the each unit.

### PART-A

- Which of the following is true about shell? CO1 (01)
  - A Shell provides you with an interface to the Unix system.
  - Shell gathers input from you and executes programs based on that input.
  - Shell is an environment in which we can run our commands, programs, and shell scripts.
  - All of the above
- Which of the following is default permission set for ordinary files? CO1 (01)
  - rw-rw-rw-
  - rw-rw-rwx
  - r-r-r-
  - rw-rw-rwx
- Which command shows some attributes of a process? CO2 (01)
  - HOME
  - ps
  - \$\$
  - pid
- \_\_\_\_\_ is a computer network administration software utility used to test the reachability of a host on an Internet Protocol network. CO2 (01)
  - finger
  - ping
  - host
  - traceroute
- Which symbol used to push the process to background? CO3 (01)
  - |
  - &
  - ^
  - \*
- What is true about pipes in Unix? CO3 (01)
  - Pipe can connect two commands together so that the output from one program becomes the input of the next program
  - To make a pipe, put a hash (#) on the command line between two commands.
  - Two or more commands connected in this way form a pipe.
  - Both A and C
- \_\_\_\_\_ option is used with the cut command for cutting fields. CO4 (01)
  - c
  - n
  - f
  - a
- 'sort' command uses \_\_\_\_\_ as the default field delimiter. CO4 (01)
  - Tab
  - Newline
  - Single space
  - One or more continuous spaces

9. What does -c operation do in grep command? CO5 (01)  
a) Matches either upper or lowercase. b) Prints only the count of matching lines.  
c) Prints the matched line and its line number. d) Prints all lines that do not match pattern.
10. Which one of the following command will be used for searching "director" in emp.lst? CO5 (01)  
a) grep "director" b) grep -v "director" emp.lst  
c) grep -director emp.lst d) grep "director" emp.lst

## PART-B

### UNIT- I

1. a) Illustrate the layered UNIX architecture with a neat diagram. CO1 (05)  
b) Write the commands to perform following operations: CO1 (03)  
i. Display line, character and word count of a file.  
ii. Copy file1 content to file2  
iii. to know the users currently logged in.
2. a) Explain two methods to change permission of files with suitable examples. CO1 (04)  
b) Explain different wildcards used in UNIX with suitable example. CO1 (04)

### UNIT- II

3. a) Analyze Unix session flow with a neat diagram. CO2 (05)  
b) Write the commands to perform following operations: CO2 (03)  
i. Sort the file, removing duplicate lines  
ii. Displays only duplicate lines  
iii. compare two files byte by byte.
4. a) Explain in detail process utilities in detail. CO2 (05)  
b) Explain 'du' command with options. CO2 (03)

### UNIT- III

5. a) Explain in detail three standard streams of UNIX. CO3 (05)  
b) Explain the types of variables in UNIX with examples. CO3 (03)
6. a) Illustrate the job states in UNIX with a flow diagram. CO3 (05)  
b) Explain the need of command substitution with an example. CO3 (03)

### UNIT- IV

7. a) Explain with examples any two UNIX commands that can be used to compare two files. CO4 (04)  
b) Write the command to perform following operations: CO4 (04)  
i. To display first five current users login.  
ii. To select only 1<sup>st</sup> and 3<sup>rd</sup> column of file1  
iii. To display two files vertically  
iv. To display number of lines in a file1.
8. a) Explain with examples. the various ways of using the 'tr' command. CO4 (05)  
b) Explain the following command with an example. CO4 (03)  
i. Tail ii. head iii. Paste.

## UNIT- V

- |    |    |                                                                                                |     |      |
|----|----|------------------------------------------------------------------------------------------------|-----|------|
| 9. | a) | Define regular expressions. Illustrate atoms and operators with the help of suitable examples. | CO5 | (05) |
|    | b) | Write the command to perform the following operations:                                         | CO5 | (03) |
|    |    | i. Select the lines from the file that have only the string UNIX.                              |     |      |
|    |    | ii. Find out if John is currently logged into the system                                       |     |      |
|    |    | iii. find a list of all files in the current directory that contain "bash".                    |     |      |
| 10 | a) | Explain the grep command with options.                                                         | CO5 | (04) |
|    | b) | Explain with examples how grep can be used to search file contents.                            | CO5 | (04) |

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